

I. SUPPLEMENTAL MATERIAL

A. Model

We use a two-dimensional tight binding model in which optical phonons are coupled to the hopping amplitude of the electrons:

$$\begin{aligned}
 H = & -t \sum_{a,i,r} \left(c_{a,i}^\dagger c_{a,i+r} + h.c. \right) \\
 & + \sum_{a,b,i,r} \left[\frac{1}{2} M \omega_0^2 (X_{ab,i}^r)^2 + \frac{1}{2M} (P_{ab,i}^r)^2 \right] \\
 & + \frac{\alpha}{\sqrt{\beta N}} \sum_{a,b,i,r} X_{ab,i}^r \left(c_{a,i}^\dagger c_{b,i+r} + h.c. + a \leftrightarrow b \right).
 \end{aligned} \tag{1}$$

The corresponding Lagrangian is

$$\begin{aligned}
 L = & \sum_{a,\nu_n} \int \frac{d^2 k}{(2\pi)^2} [i\nu_n - \xi_{\mathbf{k}}] c_a^\dagger(\mathbf{k}, \nu_n) c_a(\mathbf{k}, \nu_n) \\
 & + \sum_{a,b,\omega_n,r} \int \frac{d^2 q}{(2\pi)^2} \left[\frac{1}{2} M \omega_0^2 + \frac{1}{2} M \omega_n^2 \right] |X_{ab}^r(\mathbf{q}, \omega_n)|^2 \\
 & + \frac{\alpha}{\sqrt{\beta N}} \sum_{a,b,\nu_n,\nu_m,r} \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} X_{ab}^r(\mathbf{k} - \mathbf{k}', \nu_n - \nu_m) \times \\
 & [c_a^\dagger(\mathbf{k}, \nu_n) c_b(\mathbf{k}', \nu_m) + a \leftrightarrow b] \times (e^{ik_r a} + e^{-ik'_r a})
 \end{aligned} \tag{2}$$

where $c_a^\dagger(\mathbf{k}, \nu_n)$ creates an electron of wavevector $\mathbf{k}$, Matsubara frequency ν_n , and flavor a ; $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu$, with $\epsilon_{\mathbf{k}} = -2t \sum_r \cos(k_r)$, $r = \hat{x}, \hat{y}$, and μ the chemical potential. $X_{ab}^r(\mathbf{q}, \omega_n)$ is the fourier transform of the phonon displacement operator which lives on the r links; the phonon has mass M and optical frequency ω_0 . α is the electron-phonon coupling parameter. This model corresponds to $g_r(\mathbf{k}, \mathbf{k}') = e^{ik_r} + e^{-ik'_r}$. In the following we take $M \rightarrow \infty$, which corresponds to the limit $\omega_0 \rightarrow 0$ while keeping K , the phonon spring constant, finite.

The current operator is then given by

$$\begin{aligned}
 J(i\omega_n) = & \\
 & 2ev_0 \sum_{a,\nu_n} \int \frac{d^2 k}{(2\pi)^2} \sin(k_x) c_a^\dagger(\mathbf{k}, \nu_n) c_a(\mathbf{k}, \nu_n + \omega_n) \\
 & + \frac{e\alpha}{\sqrt{\beta N}} \frac{v_0}{t} \sum_{a,b,\nu_n,\nu_m} \int \frac{d^d k}{(2\pi)^d} \frac{d^d k'}{(2\pi)^d} X_{ab}^r(\mathbf{k} - \mathbf{k}', \nu_n - \nu_m) \\
 & [c_a^\dagger(\mathbf{k}, \nu_n) c_b(\mathbf{k}', \nu_m + \omega_n) + a \leftrightarrow b] \frac{1}{i} (e^{ik_r} - e^{-ik'_r}) \\
 & \equiv J_r^0 + J_r^1,
 \end{aligned} \tag{3}$$

with $v_0 = \delta t$, δ being the lattice spacing.

B. Single electron properties

To leading order in N^{-1} , the self energy is given by the self consistency equation:

$$\Sigma(\mathbf{k}, \omega) = \frac{\alpha^2 T}{K} \sum_r \int \frac{d^2 k'}{(2\pi)^2} \frac{[2 + 2 \cos(k_r + k'_r)]}{\omega - \xi_{\mathbf{k}'} - \Sigma(\mathbf{k}', \omega)} \tag{4}$$

Since $\xi_{\mathbf{k}}$ and $\Sigma(\mathbf{k}, \omega)$ are even functions of $\mathbf{k}$, it is possible to simplify

$$\begin{aligned}\Sigma(\mathbf{k}, \omega) &= \frac{\alpha^2 T}{K} \sum_r \int \frac{d^2 k'}{(2\pi)^2} \frac{[2 + 2 \cos(k_r) \cos(k'_r)]}{\omega - \xi_{\mathbf{k}'} - \Sigma(\mathbf{k}', \omega)} \equiv \Sigma_I(\omega, T) + \Sigma_{II}(\omega, T) \sum_r \cos(k_r), \text{ with} \\ \Sigma_I(\omega, T) &= \frac{4\lambda T}{\nu} \int \frac{d^2 k'}{(2\pi)^2} \frac{1}{\omega - \xi_{\mathbf{k}'} - \Sigma(\mathbf{k}', \omega)} = \\ &= \frac{4\lambda T}{\nu} \int d^d k' \frac{1}{\omega - \xi_{\mathbf{k}'} - \Sigma_I(\omega, T) - \Sigma_{II}(\omega, T) \sum_r \cos(k'_r)} \\ \text{and } \Sigma_{II}(\omega, T) &= \frac{2\lambda T}{\nu} \int \frac{d^2 k'}{(2\pi)^2} \frac{\cos(k'_x)}{\omega - \xi_{\mathbf{k}'} - \Sigma(\mathbf{k}', \omega)} = \\ &= \frac{2\lambda T}{\nu} \int \frac{d^2 k'}{(2\pi)^2} \frac{\cos(k'_x)}{\omega - \xi_{\mathbf{k}'} - \Sigma_I(\omega, T) - \Sigma_{II}(\omega, T) \sum_r \cos(k'_r)}.\end{aligned}\tag{5}$$

Equation 10 in the paper was solved by obtaining Σ_I and Σ_{II} iteratively for each ω , and demanding the chemical potential μ satisfy

$$n = \int \frac{d^2 k}{(2\pi)^2} \int_{-\infty}^{\mu} \frac{d\omega}{2\pi} \text{Im} \frac{1}{\omega + (2t - \Sigma_{II}(\omega))(\cos(k_x) + \cos(k_y)) - \Sigma_I(\omega)}.\tag{6}$$

At high temperatures satisfying $\lambda T \gg \Lambda$, we can neglect $\epsilon_{\mathbf{k}}$ relative to the T -dependent chemical potential and to the self energy; this will be shown to be self consistent. In this case,

$$\begin{aligned}\Sigma(\mathbf{k}, \omega) &= \frac{\alpha^2 T}{K} \sum_r \int \frac{d^2 k'}{(2\pi)^2} \frac{[2 + 2 \cos(k_r + k'_r)]}{\omega + \mu - \Sigma(\mathbf{k}', \omega)} \\ n &= \int \frac{d^2 k}{(2\pi)^2} \int_{-\infty}^{\mu} \frac{d\omega}{2\pi} \text{Im} \frac{1}{\omega - \Sigma(\mathbf{k}, \omega)}.\end{aligned}\tag{7}$$

We define the dimensionless quantities

$$\begin{aligned}\tilde{\omega} &= \omega / \sqrt{\lambda T / \nu} \\ \tilde{\Sigma}(\mathbf{k}, \omega) &= \Sigma(\mathbf{k}, \omega) / \sqrt{\lambda T / \nu} \\ \tilde{\mu}_0 &= \mu(T) / \sqrt{\lambda T / \nu}.\end{aligned}\tag{8}$$

They satisfy the temperature independent equations in Eq. 11 in the paper, and depend only on the density. $\tilde{\mu}_0$ and $\tilde{\Sigma}(\mathbf{k}, \omega \ll \sqrt{\lambda T / \nu})$ are found to be numbers of order one, rendering our approximation consistent.

C. σ^{00}

Next, we calculate the J^{00} renormalized vertex. In the calculations of the conductivity, we calculate σ_{xx} , and therefore only consider $g_{r=x}(\mathbf{k}, \mathbf{k}')$. In addition, we set the electron charge $e = 1$ for simplicity. As shown in Fig. 1, only ladder diagrams contribute to the vertex to lowest order in $1/N$. The (x -direction) vertex function $\Gamma(\mathbf{k}, \nu_n, \nu_n + \omega_n)$

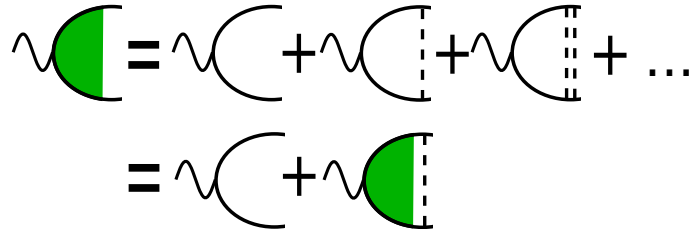


FIG. 1. (Color online.) The J^{00} vertex. Only ladder diagrams contribute to the vertex to lowest order in $1/N$.

satisfies

$$\begin{aligned}\Pi^{00}(i\omega_n) &= N \int \frac{d^2k}{(2\pi)^2} \frac{1}{\beta} \sum_n G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \Gamma(\mathbf{k}, \nu_n, \nu_n + \omega_n) \times 2t \sin(k_x a), \text{ with} \\ \Gamma(\mathbf{k}, \nu_n, \nu_n + \omega_n) &= 2t \sin(k_x a) \\ &+ \frac{2\alpha^2}{K} \int \frac{d^2k'}{(2\pi)^2} \frac{1}{\beta} \sum_m [1 + \cos(k_x + k'_x)] D(\mathbf{k} - \mathbf{k}', \nu_n - \nu_m) G(\mathbf{k}', \nu_m) G(\mathbf{k}', \nu_m + \omega_n) \Gamma(\mathbf{k}', \nu_m, \nu_m + \omega_n)\end{aligned}\quad (9)$$

Performing the Matsubara summation, to lowest order in ω_0/T only the pole at the phonon propagator is taken into account, and we are left with the self consistent equation

$$\begin{aligned}\Gamma(\mathbf{k}, \nu_n, \nu_n + \omega_n) &= 2t \sin(k_x a) \\ &+ \frac{2\lambda T}{\nu} \int \frac{d^2k'}{(2\pi)^2} [1 + \cos(k_x + k'_x)] G(\mathbf{k}', \nu_n) G(\mathbf{k}', \nu_n + \omega_n) \Gamma(\mathbf{k}', \nu_n, \nu_n + \omega_n)\end{aligned}\quad (10)$$

Using the ansatz $\Gamma(\mathbf{k}, \nu_m, \nu_m + \omega_n) = \gamma(\nu_n, \nu_n + \omega_n) \sin(k_x)$, we get (using the fact that $G(\mathbf{k}, \nu_n)$ is an even function of k_x)

$$\begin{aligned}\gamma(\nu_n, \nu_n + \omega_n) &= 2t - \frac{2\lambda T}{\nu} \gamma(\nu_n, \nu_n + \omega_n) \int \frac{d^2k}{(2\pi)^2} G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \sin^2(k_x) \\ &= \frac{2t}{1 + \frac{2\lambda T}{\nu} F(\nu_n, \nu_n + \omega_n)},\end{aligned}\quad (11)$$

with

$$F(\nu_n, \nu_n + \omega_n) = \int \frac{d^2k}{(2\pi)^2} G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \sin^2(k_x) \quad (12)$$

We now calculate $\Pi^{00}(i\omega_n)$:

$$\begin{aligned}\Pi^{00}(i\omega_n) &= N \int \frac{d^2k}{(2\pi)^2} \frac{1}{\beta} \sum_n G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \times 2t \sin(k_x) \Gamma(\mathbf{k}, \nu_n, \nu_n + \omega_n) \\ &= N(2t)^2 \frac{1}{\beta} \sum_n \frac{F(\nu_n, \nu_n + \omega_n)}{1 + \frac{2\lambda T}{\nu} F(\nu_n, \nu_n + \omega_n)} \equiv N \frac{(2t)^2}{\beta} \sum_n P(\nu_n, \nu_n + \omega_n)\end{aligned}$$

We perform the Matsubara summation following [1], using the usual contour integral method in the complex plane. The complex function $G(\mathbf{k}, z)$, which satisfies (δ is an infinitesimal)

$$\begin{aligned}G(\mathbf{k}, z = \omega + i\delta) &= G^R(\mathbf{k}, \omega) \\ G(\mathbf{k}, z = \omega - i\delta) &= G^A(\mathbf{k}, \omega) = G^R(\mathbf{k}, \omega)^*\end{aligned}\quad (13)$$

has a branch cut on the real axis. Therefore, $P(z, z + \omega_n)$ has branch cuts at $\Re[z] = 0, -\omega_n$. Performing this integral (and using the fact that $n_F(\epsilon - i\omega_n) = n_F(\epsilon)$) results in

$$\begin{aligned}\frac{\Pi^{00}(i\omega_n)}{(2t)^2 N} &= \frac{1}{\beta} \sum_n P(i\nu_n, i\nu_n + i\omega_n) = \\ &- \int \frac{d\epsilon}{2\pi i} n_F(\epsilon) [P(\epsilon + i\delta, \epsilon + i\omega_n) - P(\epsilon - i\delta, \epsilon + i\omega_n)] - \int \frac{d\epsilon}{2\pi i} n_F(\epsilon) [P(\epsilon - i\omega, \epsilon + i\delta) - P(\epsilon - i\omega, \epsilon - i\delta)]\end{aligned}\quad (14)$$

Therefore, using the fact that $T \ll \mu$, and thus $dn_F(\epsilon)/d\epsilon \approx -\delta(\epsilon)$,

$$\sigma^{00} = -\frac{1}{\omega} \lim_{\omega \rightarrow 0} \text{Im} \Pi^{00}(i\omega_n \rightarrow \omega + i\delta) = N \frac{(2t)^2}{2\pi} [P(-i\delta, i\delta) - \Re[P(i\delta, i\delta)]], \quad (15)$$

where

$$\begin{aligned}
P(-i\delta, i\delta) &= \frac{F(-i\delta, i\delta)}{1 + \frac{2\lambda T}{\nu} F(-i\delta, i\delta)}, \\
P(i\delta, i\delta) &= \frac{F(i\delta, i\delta)}{1 + \frac{2\lambda T}{\nu} F(i\delta, i\delta)}, \\
F(-i\delta, i\delta) &= \int \frac{d^2 k}{(2\pi)^2} G^A(\mathbf{k}, 0) G^R(\mathbf{k}, 0) \sin^2(k_x), \\
F(i\delta, i\delta) &= \int \frac{d^2 k}{(2\pi)^2} G^R(\mathbf{k}, 0) G^R(\mathbf{k}, 0) \sin^2(k_x);
\end{aligned} \tag{16}$$

these are easily computed once we have $\Sigma_I(0)$ and $\Sigma_{II}(0)$.

D. σ^{01}

The 01 current-current correlation function is given, to lowest order in $1/N$, by

$$\begin{aligned}
\Pi^{01}(i\omega_n) &= -i\alpha^2 N \frac{1}{\beta^2} \sum_{n,m} \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} D(\mathbf{k} - \mathbf{k}') \times \\
&G(\mathbf{k}', \nu_m) G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \Gamma(\mathbf{k}, \nu_n, \nu_n + \omega_n) \left(e^{ik_x} - e^{-ik'_x} \right) \left(e^{-ik_x} + e^{ik'_x} \right).
\end{aligned} \tag{17}$$

We perform the summation over ν_m ; to lowest order in ω_0/T , this gives us (using the fact the all Green's functions are even in k_x , and inserting the vertex function 11):

$$\begin{aligned}
\Pi^{01}(i\omega_n) &= N(2t)^2 \frac{\lambda T}{\nu t} \frac{1}{\beta} \sum_n \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} \sin^2(k_x) \times \\
&G(\mathbf{k}', \nu_n) G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \frac{1}{1 + \frac{2\lambda T}{\nu} F(\nu_n, \nu_n + \omega_n)} \equiv N(2t)^2 \frac{\lambda T}{\nu t} \frac{1}{\beta} \sum_n R(\nu_n, \nu_n + \omega_n).
\end{aligned} \tag{18}$$

Note that to this order, the J_1 vertex is not renormalized.

Just as in the discussion above Eq. 14, the complex function $R(z, z + \omega_n)$ has branch cuts at $\Re[z] = 0, -\omega_n$; performing the Matsubara summation as in Eq. 14, we get

$$\begin{aligned}
\Pi^{01}(i\omega_n) / \left[N(2t)^2 \frac{\lambda T}{4\nu t} \right] &= \frac{1}{\beta} \sum_n R(i\nu_n, i\nu_n + i\omega_n) = \\
&- \int \frac{d\epsilon}{2\pi i} n_F(\epsilon) [R(\epsilon + i\delta, \epsilon + i\omega_n) - R(\epsilon - i\delta, \epsilon + i\omega_n)] - \int \frac{d\epsilon}{2\pi i} n_F(\epsilon) [R(\epsilon - i\omega, \epsilon + i\delta) - R(\epsilon - i\omega, \epsilon - i\delta)]
\end{aligned} \tag{19}$$

Therefore, in a procedure similar to that of Eq. 14, we get

$$\sigma^{01} = N \frac{\lambda T}{\nu t} \frac{(2t)^2}{2\pi} \Re [R(-i\delta, i\delta) - R(i\delta, i\delta)], \tag{20}$$

with

$$\begin{aligned}
R(-i\delta, i\delta) &= \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} \sin^2(k_x) \frac{G^A(\mathbf{k}', 0) G^A(\mathbf{k}, 0) G^R(\mathbf{k}, 0)}{1 + \frac{\lambda T}{2\nu} F(-i\delta, i\delta)} = P(-i\delta, i\delta) \int \frac{d^2 k'}{(2\pi)^2} G^A(\mathbf{k}', 0) \\
R(i\delta, i\delta) &= \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} \sin^2(k_x) \frac{G^R(\mathbf{k}', 0) G^R(\mathbf{k}, 0) G^R(\mathbf{k}, 0)}{1 + \frac{\lambda T}{2\nu} F(i\delta, i\delta)} = P(i\delta, i\delta) \int \frac{d^2 k'}{(2\pi)^2} G^R(\mathbf{k}', 0),
\end{aligned} \tag{21}$$

where the P functions are defined in Eq. 16. Again, these can be computed easily once the self energy is obtained.

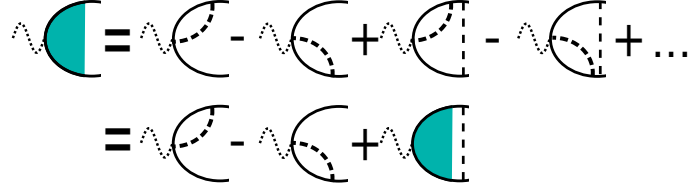


FIG. 2. (Color online.) The J^{11} vertex.

E. σ^{11}

Finally, we turn to compute the conductivity in the 11 channel. The contribution of the third diagram in Fig. ??, which we term σ_a^{11} , is given by substituting in Eq. 7 in the paper

$$g_x(\mathbf{k}, \mathbf{k}') = e^{ik_x} + e^{-ik'_x} \quad (22)$$

to get

$$\sigma_a^{11} = N \frac{\lambda T}{2\pi\nu} \int \frac{d^2k}{(2\pi)^2} \frac{d^2k'}{(2\pi)^2} [1 - \cos(k_x) \cos(k'_x)] A(\mathbf{k}, 0) A(\mathbf{k}', 0). \quad (23)$$

The calculation of diagram (D) of Fig. 2 in the paper, which we denote σ_b^{11} , includes the renormalized J_1 vertex. This vertex, which is schematically described in Fig. 2, obeys the self-consistent equation (in a manner completely analogous to the J_0 vertex)

$$\begin{aligned} \Pi_b^{11} &= N \frac{2\lambda T}{\nu} \int \frac{d^2k d^2k'}{(2\pi)^4} \frac{1}{\beta} \sum_n \sin(k_x + k'_x) G(\mathbf{k}, \nu_n) G(\mathbf{k}, \nu_n + \omega_n) \\ &\quad \times [G(\mathbf{k}', \nu_n) - G(\mathbf{k}', \nu_n + \omega_n)] \Delta(\mathbf{k}, \nu_n, \nu_n + \omega_n), \text{ with} \\ \Delta(\mathbf{k}, \nu_n, \nu_n + \omega_n) &= \frac{2\lambda T}{\nu} \int \frac{d^2k'}{(2\pi)^2} \sin(k_x + k'_x) [G(\mathbf{k}', \nu_n + \omega_n) - G(\mathbf{k}', \nu_n)] \\ &\quad + \frac{2\lambda T}{\nu} \int \frac{d^2k'}{(2\pi)^2} [1 + \cos(k_x + k'_x)] G(\mathbf{k}', \nu_n + \omega_n) G(\mathbf{k}', \nu_n) \Delta(\mathbf{k}', \nu_n, \nu_n + \omega_n) \end{aligned} \quad (24)$$

Using the fact that $G(\mathbf{k}, \nu_n)$ is an even function of k_x , and using the ansatz

$$\Delta(\mathbf{k}, \nu_n, \nu_n + \omega_n) = \frac{2\lambda T}{\nu} \delta(\nu_n, \nu_n + \omega_n) \sin(k_x), \quad (25)$$

we arrive at the result

$$\delta(\nu_n, \nu_n + \omega_n) = \frac{F_1(\nu_n, \nu_n + \omega_n)}{1 + \frac{2\lambda T}{\nu} F(\nu_n, \nu_n + \omega_n)}, \quad (26)$$

with $F_1(\nu_n, \nu_n + \omega_n) = \int \frac{d^2k}{(2\pi)^2} \cos(k_x) [G(\mathbf{k}, \nu_n + \omega_n) - G(\mathbf{k}, \nu_n)]$ and $F(\nu_n, \nu_n + \omega_n)$ given by Eq. 12.

Performing the Matsubara summation results in

$$\sigma_b^{11} = -4 \left(\frac{\lambda T}{\nu} \right)^2 \frac{1}{2\pi} \frac{F_1^2(-i\delta, i\delta) F(-i\delta, i\delta)}{1 + \frac{2\lambda T}{\nu} F(-i\delta, i\delta)} \quad (27)$$

with $F_1(-i\delta, i\delta) = \int \frac{d^2k}{(2\pi)^2} \cos(k_x) [G^R(\mathbf{k}, 0) - G^A(\mathbf{k}, 0)]$.

F. Optical conductivity

Using eqns.14 and 19, we get

$$\begin{aligned}\sigma^{00}(\omega) &= -\frac{N}{\omega} \text{Im}\Pi^{00}(i\omega_n \rightarrow \omega + i\delta) = N \frac{(2t)^2}{2\pi} \Re \int_{-\omega}^0 d\epsilon [P(\epsilon - i\delta, \epsilon + \omega + i\delta) - P(\epsilon + i\delta, \epsilon + \omega + i\delta)], \\ \sigma^{01}(\omega) &= -\frac{N}{\omega} \text{Im}\Pi^{01}(i\omega_n \rightarrow \omega + i\delta) = N \frac{\lambda T}{\nu t} \frac{(2t)^2}{2\pi} \Re \int_{-\omega}^0 d\epsilon [R(\epsilon - i\delta, \epsilon + \omega + i\delta) - R(\epsilon + i\delta, \epsilon + \omega + i\delta)],\end{aligned}\quad (28)$$

while

$$\sigma^{11}(\omega) = \frac{\lambda T a^2}{2\pi\nu} \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} [1 + \cos(k_x) \cos(k'_x)] \int_{-\omega}^0 d\epsilon A(\mathbf{k}, \epsilon) A(\mathbf{k}', \epsilon + \omega); \quad (29)$$

where we have again used the fact that $T \ll E_F$.

G. Numerical simulation

In the numerical simulation, we consider a two dimensional tight binding model of size $L \times L$, and N flavors of electrons. The single-particle Hamiltonian is given by

$$\begin{aligned}H &= H_0 + H_{el-ph} \\ H_0 &= -t \sum_{i,j,\alpha,\eta=\pm 1} (|i, j, \alpha\rangle \langle i + \eta, j, \alpha| + |i, j, \alpha\rangle \langle i, j + \eta, \alpha|) - t' \sum_{i,j,\alpha,\eta,\eta'=\pm 1} |i, j, \alpha\rangle \langle i + \eta, j + \eta', \alpha| \\ H_{el-ph} &= -\frac{\alpha}{\sqrt{N}} \sum_{i,j,\alpha} [X_{i,j,\alpha,\beta}^1 |i, j, \alpha\rangle \langle i + 1, j, \beta| + X_{i,j,\alpha,\beta}^2 |i, j, \alpha\rangle \langle i, j + 1, \beta|] + h.c.\end{aligned}\quad (30)$$

where $|i, j, \alpha\rangle$ is the state with an electron of flavor $\alpha \in [1, N]$ at site i, j . t is the nearest neighbor hopping parameter, while t' represents next nearest neighbor hopping. α is the electron-phonon coupling strength, and $X_{\alpha,\beta,i,j}^r$ are the phonon modes at each site; they are characterized by the spring constant K . We choose $t' = -t$ and scale all energies by t . In addition, we work at filling $n = 0.4$; these were chosen in order to minimize the susceptibility to lattice instabilities. Periodic boundary conditions were imposed.

The phonons are treated as classical fields and have no dynamics. This is justified in the limit $T \gg \omega_0$, where the quantum mechanical nature of the phonons is insignificant. However, we do consider the backaction of the electrons on the phonons. In our Monte Carlo simulation, at each step a single $X_{\alpha,\beta,i,j}^r$ is changed, and the resulting free energy is computed by

$$F = \frac{1}{2} K \sum_{i,j,\alpha,\beta,r} (X_{i,j,\alpha,\beta}^r)^2 - T \sum_n \log \left(1 + e^{(-\epsilon_n - \mu)/T} \right), \quad (31)$$

where ϵ_n are the $L \times L \times N$ eigenvalues of the single-particle Hamiltonian for the given phonon configuration, and μ is the chemical potential obtained by demanding constant filling. The ϵ_n 's are obtained by exact diagonalization. The fact that the electrons modify the phonon configuration via the second term in 31, allows us to observe lattice (Pierels) instabilities which occurs at strong couplings and low temperatures.

Each frozen phonon configuration represents a free electron system. It is therefore possible to calculate the conductivity by defining the (x -direction) single particle current operator

$$J_x = -i \left[-t \sum_{i,j,\alpha,\eta=\pm 1} \eta |i, j, \alpha\rangle \langle i + \eta, j, \alpha| - \frac{\alpha}{\sqrt{N}} \sum_{i,j,\alpha,\beta,\eta=\pm 1} X_{i,j,\alpha,\beta}^1 \eta |i, j, \alpha\rangle \langle i + \eta, j, \beta| \right], \quad (32)$$

and calculating the optical conductivity as

$$\sigma(\omega) = \frac{\pi}{L^2 \omega} \sum_{n,n'} |\langle n | J_x | n' \rangle|^2 [n_F(\epsilon_n) - n_F(\epsilon_{n'})] \delta(\omega + \epsilon_n - \epsilon_{n'}), \quad (33)$$

with $|n\rangle$ the eigenstate corresponding to ϵ_n ; again, both are obtained by exact diagonalization. We broaden the δ -

function by using

$$\delta(\epsilon) \approx -\pi \text{Im} \frac{1}{\epsilon + 0.01i} \quad (34)$$

In order to reduce finite size effects, we insert a flux quantum of 2π through the system. The effect of this flux is to break the point group and translation symmetries of the problem, which facilitates the convergence to the thermodynamic results [2]. A gauge choice that describes a single flux quantum through a square lattice with a first and second neighbor hopping appears in Ref. [3]. For $N = 8$ we use an 8×8 lattice, and a finite size gap appears in the form of a dip in $\sigma(\omega \rightarrow 0)$, as shown in Fig. 3. We ascertain that this dip vanishes as L is increased, and is not due to an instability, such as a Pierels gap. We therefore approximate the conductivity as $\sigma_{dc} \approx \sigma(\omega = 0.0083\Lambda)$.

The system sizes we use are $L = 8$ for $N = 8$, $L = 10$ for $N = 6$, $L = 12$ for $N = 4$, and $L = 17$ for $N = 2$. In each simulation, 80 sweeps of the entire system were performed for thermalization, and 900 calculations of $\sigma(\omega)$ were averaged. We choose the coupling α such that the dimensionless parameter

$$\tilde{c} \equiv \frac{\alpha^2}{Kt} = 3, \quad (35)$$

which is large enough to observe saturation at $T \ll E_F$, while small enough so that there is no Pierels transition at the temperatures we consider.

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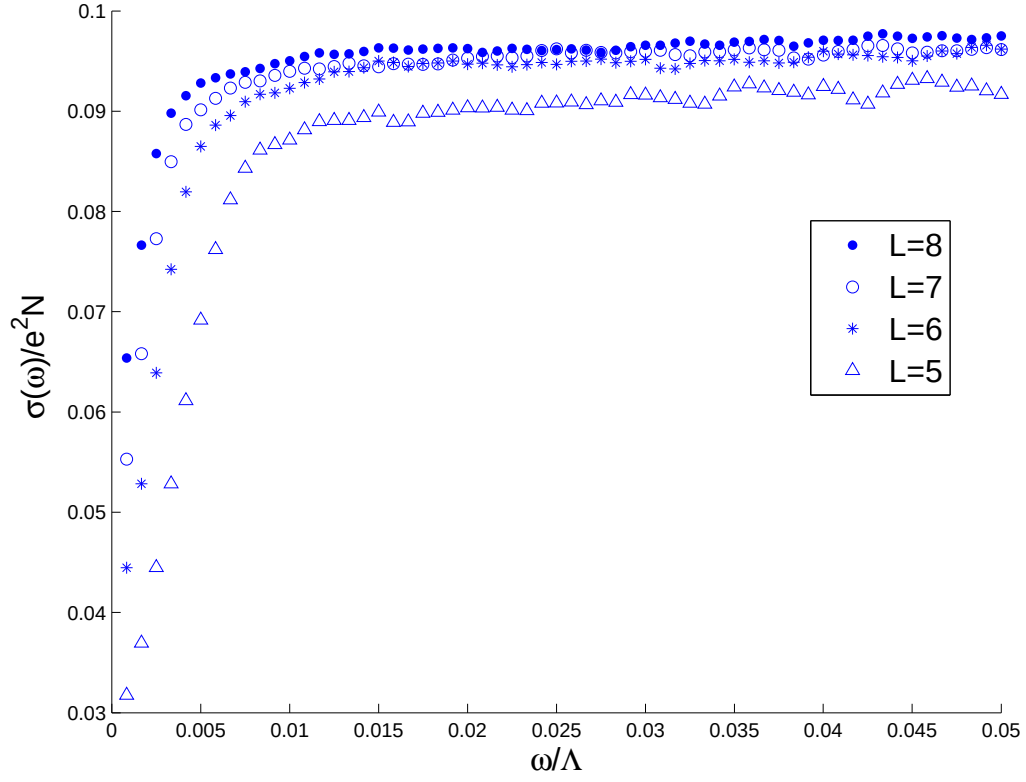


FIG. 3. (Color online.) The optical conductivity for $N = 8$ and several system sizes. The dip in the optical conductivity at low frequencies is due to a finite size gap, as shown by its decrease with increasing system size.